

Inverse of a Matrix and the Matrix Equation $A\vec{x} = B$

Example

Consider the system of linear equations

$$\begin{aligned} 2x - 7y &= 3 \\ 5x - 18y &= 8 \end{aligned}$$

and use the concept of **the inverse of a matrix** to solve the system.

We can write the matrix eq. $A\vec{x} = \vec{b}$ as follows: $A\vec{x} = \begin{bmatrix} 2 & -7 \\ 5 & -18 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 8 \end{bmatrix}}_{\vec{b}}$

If A is invertible then: $\vec{A}^{-1}(A\vec{x}) = \vec{A}^{-1}(\vec{b})$

$$(\vec{A}^{-1}A)\vec{x} = \vec{A}^{-1}\vec{b} \rightarrow I\vec{x} = \vec{A}^{-1}\vec{b} \rightarrow \boxed{\vec{x} = \vec{A}^{-1}\vec{b}}$$

$$\vec{A}^{-1} = \begin{bmatrix} 18 & -7 \\ 5 & -2 \end{bmatrix} \rightarrow \vec{x} = \begin{bmatrix} 18 & -7 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 8 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

$x = -2$ and $y = -1$ is a unique solution for this system.

How do we know $\vec{x}$ is a unique solution? Let's assume that there exists more than one solution. For example, if $\vec{w}$ is another solution then we can repeat the above process to show that $\vec{w} = \vec{A}^{-1}\vec{b}$. Also, since $\vec{x} = \vec{A}^{-1}\vec{b}$, then $\vec{x} = \vec{w}$, which means $\vec{x}$ and $\vec{w}$ are the same.

Theorem (Inverses of Transposes and Products)

- 1 If A is an invertible matrix, then $(A^T)^{-1} = (A^{-1})^T$.
- 2 If A and B are invertible matrices, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
- 3 If $A_1, A_2, \dots, A_k$ are invertible, then $A_1A_2 \cdots A_k$ is invertible and $(A_1A_2 \cdots A_k)^{-1} = A_k^{-1}A_{k-1}^{-1} \cdots A_2^{-1}A_1^{-1}$.

- 1 If A is an invertible matrix, then $AA^{-1} = I$, which means

$$\begin{aligned} & (AA^{-1})^T = I^T \longrightarrow (A^{-1})^T A^T = I \\ \text{Also, Since } A^{-1}A = I & \longrightarrow (A^{-1}A)^T = A^T (A^{-1})^T = I \end{aligned} \quad \left. \vphantom{\begin{aligned} & (AA^{-1})^T = I^T \longrightarrow (A^{-1})^T A^T = I \\ & (A^{-1}A)^T = A^T (A^{-1})^T = I \end{aligned}} \right\} \Rightarrow (A^{-1})^T \text{ is inverse of } A^T \text{ or} \\ & (A^{-1})^T = (A^T)^{-1}. \end{aligned}$$

Exercise: Prove ② and ③.

Elementary Matrices

Definition

An **elementary matrix** is a matrix obtained from an identity matrix by performing a **single** elementary row operation.

Reminder: Elementary Row Operations

- 1 Interchange two rows.
- 2 Multiply a row by a nonzero number.
- 3 Add a (nonzero) multiple of one row to a different row.

Example

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Let us examine the effect of left multiplication of A by the above elementary matrices.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{bmatrix} \quad EA = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 4 \\ 3 & 3 \\ 2 & 2 \end{bmatrix}$$
$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = E$$

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-2R_3} F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$FA = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ -6 & -6 \\ 4 & 4 \end{bmatrix}$$

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-3R_1 + R_3} G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$GA = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 0 \\ 4 & 4 \end{bmatrix}$$

Observation: Left multiplication of a matrix by the elementary matrices induces the same row operation used to obtain them from the identity matrix.

Inverses of Elementary Matrices

Example

Without using the matrix inversion algorithm, find the inverse of the elementary matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Hint. What row operation can be applied to G to transform it to I_4 ?

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\text{row operation}]{-3R_1+R_3} G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\text{Inverse row operation}]{+3R_1+R_3} G^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ +3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

observation : For finding inverse of an elementary matrix E , we can apply to I the inverse of the row operation applied to I to obtain E . This will be more efficient than the matrix inversion algorithm.

A Matrix as a Product of Elementary Matrices

Example

Express $A = \begin{bmatrix} 1 & 2 & -4 \\ -3 & -6 & 13 \\ 0 & -1 & 2 \end{bmatrix}$ as a product of elementary matrices and, if possible, find A^{-1} .

We need to keep track of the elementary row operations performed in the matrix inversion algorithm.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ -3 & -6 & 13 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 + 3R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 & 1 & 0 \end{array} \right] \xrightarrow{-R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & 1 & 0 \end{array} \right]$$

$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -4 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & 1 & 0 \end{array} \right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & -2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & 1 & 0 \end{array} \right] \xrightarrow{R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & 1 & 0 \end{array} \right]$$

$E_4 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

$$(E_5 E_4 E_3 E_2 E_1) A = I \longrightarrow (E_5 E_4 E_3 E_2 E_1) \underbrace{A \bar{A}^{-1}}_I = I \bar{A}^{-1}$$

$$\longrightarrow \bar{A}^{-1} = E_5 E_4 E_3 E_2 E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 6 & 2 & -1 \\ 3 & 1 & 0 \end{bmatrix}$$

$$\bar{A}^{-1} = E_5 E_4 E_3 E_2 E_1 \longrightarrow A = (E_5 E_4 E_3 E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} E_5^{-1}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 20 \\ 0 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$